

BEM-C302

SEMESTER EXAMINATION- DECEMBER 2024

CLASS:B.TECH SEMESTER:III

SUBJECT:MATHEMATICS

PAPER TITLE:ENGINEERING MATHEMATICS-III

Time: 3 hour

Max. Marks: 70

Note: Question Paper is divided into two sections: **A and B**. Attempt both the sections as per given instructions.

SECTION-A (SHORT ANSWER TYPE QUESTIONS)		CO	BL
Instructions: Answer any <i>five</i> questions in about 150 words each. Each question carries six marks. (5 X 6 = 30 Marks)			
Q-1	Evaluate $L\{f(t)\}$ and $L\{f'(t)\}$, where $f(t) = \frac{\sin t}{t}$	CO1	L5
Q-2	Express the following function in terms of the Heaviside's unit step function and determine its Laplace transform $f(t) = \begin{cases} e^{-t}, & 0 < t < 3 \\ 0, & t > 3 \end{cases}$	CO1	L2
Q-3	Find Fourier Cosine transform of the function $f(x)$ if $f(x) = \begin{cases} \cos x, & 0 \leq x \leq a \\ 0, & x > a \end{cases}$	CO2	L1
Q-4	Express the function $f(x) = \begin{cases} 1, & x \leq 1 \\ 0, & x > 1 \end{cases}$ as a Fourier integral and hence evaluate $\int_0^\infty \frac{\sin \lambda \cos \lambda x}{\lambda} d\lambda$	CO2	L2
Q-5	Find the inverse Z-transform of $f(z) = \frac{3z^2 - 18z}{(z-2)(z-3)(z-4)}$	CO3	L1
Q-6	State the initial value theorem and using of it determine $f(0)$ and $f(1)$ when $F(z) = \frac{2z^2 + 3z + 12}{(z-1)^4}$	CO3	L3
Q-7	Evaluate the value of integral $\int_0^{2+i} (\bar{z})^2 dz$ along the real axis from $z = 0$ to $z = 2$ and then along a line parallel to imaginary axis from $z = 2$ to $z = 2 + i$	CO4	L5
Q-8	Determine the poles and residue at each pole of the following function $f(z) = \frac{z^2}{(z-1)^2(z+2)}$ and hence evaluate $\int_C \frac{z^2}{(z-1)^2(z+2)} dz$ where $ z =3$	CO4	L5
Q-9	Find the real root of the equation $x^2 - 5x + 2 = 0$ correct upto 4 decimal places by using Newton-Raphson's Method.	CO5	L3
Q-10	Apply iteration method to approximate the real root of the equation $5x = \sin x + 2$ correct upto 3 decimal places.	CO5	L3

SECTION-B (LONG ANSWER TYPE QUESTIONS)		CO	BL
Instructions: Answer any <i>four</i> questions in detail. Each question carries 10 marks. (4 X 10 = Marks)			
Q-1	Use Laplace transform to solve the differential equation $y'' - 2y' + y = e^t$ where $y(0) = 2, y'(0) = -1$	CO1	L3
Q-2	State Convolution theorem and by using it, find the inverse Laplace transform of $\frac{1}{s^2(s+1)^2}$	CO1	L3
Q-3	Solve the equation $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$, subject to the conditions i) $u = 0$ when $x = 0, t > 0$ ii) $u = \begin{cases} 1, & 0 < x < 1 \\ 0, & x \geq 1 \end{cases}$ when $t = 0$ iii) $u(x, t)$ is bounded.	CO2	L3
Q-4	Using Z-transform, solve the following difference equation $y_{n+2} + 4y_{n+1} + 3y_n = 3^n$, where $y(0) = 0$ and $y(1) = 1$	CO3	L3
Q-5	Define a harmonic function. If $u(x, y) = x^4 - 6x^2y^2 + y^4$, then show that $u(x, y)$ is harmonic function. Also find $v(x, y)$ if $f(z) = u + iv$ is an analytic function and express $f(z)$ in terms of z .	CO4	L2
Q-6	Using Contour integration technique, evaluate the real integral $\int_0^{2\pi} \frac{\sin^2 \theta}{5 - 4\cos \theta} d\theta$	CO4	L5
Q-7	Compute the real root of the equation $\cos x - xe^x = 0$ using the secant method correct upto 4 decimal places.	CO5	L3
Q-8	Find an iterative formula to find $\sqrt[3]{N}$ using Newton – Raphson's method, where N is a positive number and hence find $\sqrt[3]{17}$ correct to 4 decimal places.	CO5	L3
